

Dual-Band Terahertz Metamaterial Absorber with Near-Unity Absorption for Explosive Identification

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Abstract—Explosive detection remains a critical challenge for global security due to the limitations of conventional screening technologies, which often suffer from destructive sampling and high false alarm rates. Terahertz (THz) spectroscopy provides a promising alternative through its ability to non-invasively probe molecular vibrational fingerprints of energetic compounds. In this work, we present a compact dual-band THz metamaterial absorber ($55 \times 55 \times 40.12 \mu\text{m}^3$) designed for highly selective explosive detection. The proposed unit cell, consisting of a concentric ring with embedded square slots on an FR-4 substrate backed by copper, achieves near-unity absorption at 0.844 THz and 1.478 THz. Electromagnetic simulations reveal strong electric and magnetic field confinement, producing sensitivities up to 351.81 GHz/RIU with absorption exceeding 99%. Comparative analysis demonstrates that the sensor reliably distinguishes between TNT, Tetryl, Semtex-H, HMX, and RDX through resonance frequency shifts, maintaining stable absorbance across analytes. The first resonance exhibits an acceptable Figure of Merit (2.75) and Q-factor (11.96), highlighting its precision for trace detection. These findings establish the proposed absorber as a robust, and label-free THz sensing platform for practical security and defense applications.

Keywords—Terahertz sensing, metamaterial, THz absorber, dual-band, explosive detection, refractive index, high sensitivity, security applications.

I. INTRODUCTION

Recent years have witnessed remarkable progress in the field of explosive detection, with innovations spanning forensic chemistry, nanotechnology, optical sensing, and field-deployable analytical methods [1]. In 2023, over 47,000 casualties were attributed to explosive weapons, with civilians accounting for more than 70% [2]. High explosives such as TNT, PETN, RDX, HMX, and Tetryl are frequently employed due to their high detonation velocity, stability, and ease of concealment, which complicates conventional security measures. Such considerations highlight the necessity of developing advanced, dependable, and selective sample identification technologies to effectively mitigate explosive-related threats across military and civilian domains [3, 4]. Conventional methods such as X-ray imaging, ion mobility spectrometry, and chromatography often suffer from destructive sampling, high false alarm rates, or limited

capability to detect explosives hidden under packaging [1, 3]. Recent covert evaluations have highlighted critical weaknesses in the ability of security agents to detect explosives. Real-world tests reveal persistent gaps in explosive detection, showing personnel sometimes fail to identify mock explosives, emphasizing need for improved procedures. Given the failure rates of conventional screening, Terahertz (THz) radiation (0.1–10 THz) has emerged as a promising detection modality, capable of non-ionizing interrogation, penetrating dielectric barriers, and revealing the characteristic vibrational fingerprints of energetic compounds. Despite these advantages, THz systems face challenges such as weak source intensity and limited interaction with target molecules. Metamaterial absorbers (MMAs) address these issues by enabling strong field confinement and near-unity absorption at selected frequencies.

Recent research has demonstrated significant advances in terahertz (THz) sensor technology for explosive detection, with THz metamaterial absorbers achieving near-unity absorption (>90–99%) at explosive-specific frequencies through engineered subwavelength resonators. Wekalao et al. [5] developed a graphene and gold metasurface-based THz sensor capable of detecting nitroglycerine, TNT, and RDX with exceptional sensitivity of 927 GHz/RIU and detection limit of 0.189 RIU⁻¹. In [6] Dhillon et al. presented an ultrathin polarization-insensitive metamaterial absorber with three discrete absorption peaks at 5.57 GHz, 7.969 GHz, and 13.44 GHz, achieving absorption intensities exceeding 97% for defense and explosive detection applications. Sharma et al. [7] designed copper-silicon based THz metamaterial perfect absorbers specifically targeting RDX detection, achieving ~100% absorption at characteristic frequencies of 0.8, 1.05, 1.3, and 1.91 THz. Additionally, Haghverdi et al. [8] developed a methane detection approach using nested graphene rings in THz wave absorbers, highlighting the versatility of metamaterial-based sensing for gas detection in healthcare applications. Building on this concept, integrating metamaterial absorbers (MMAs) into THz detection platforms can provide compact, polarization-insensitive, and highly selective solutions for real-world explosive detection.

Building upon these advancements, this work introduces a compact, dual-band THz metamaterial absorber

($55 \times 55 \times 40.12 \mu\text{m}^3$) specifically designed to detect multiple explosives. The device operates on the principle of resonance frequency shifting caused by the analyte refractive index (RI). Through rigorous design optimization and computational validation, the absorber offers a highly sensitive, non-contact, and non-destructive platform, demonstrating significant potential for deployment in advanced security screening applications.

II. DEVELOPMENT METHODOLOGY

The proposed terahertz (THz) metamaterial absorber is designed to detect analytes through resonance frequency shifts, as illustrated in Fig. 1. The unit cell consists of concentric rings with embedded square slots, fabricated on a dielectric substrate backed by a metallic ground plane to suppress transmission and enhance absorption. This configuration supports multiple resonances in the THz regime, where strong field confinement and plasmonic coupling ensure high absorption efficiency. Electromagnetic simulations are performed across 0.1–2.5 THz to obtain absorbance, reflectance, and transmittance spectra, from

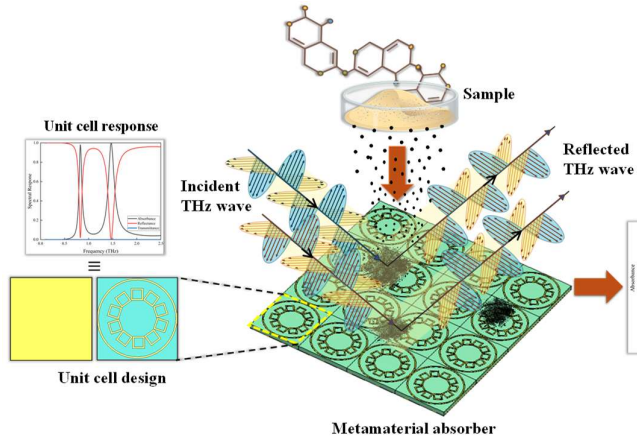


Fig. 1: Process workflow of THz metamaterial absorber for detection explosive substance.

which the baseline resonance peaks are defined as the reference response. For sensing applications, when an analyte, such as a biochemical or powdered sample, is deposited on the absorber surface, the effective dielectric environment around the resonators changes. The presence of the analyte modifies the effective refractive index near the resonator, thereby perturbing the localized field distribution and producing measurable shifts in the resonance frequencies compared to the reference spectrum. This shift provides the basis for material detection and characterization, where the designed geometry ensures both high sensitivity and strong absorption performance in the THz regime. The sensing mechanism therefore relies on tracking the frequency displacement (Δf), where sharper high-Q resonances enable greater sensitivity and precision.

III. DESIGN INVESTIGATION

A. Structural Design of Unit Cell

The proposed terahertz (THz) metamaterial absorber is constructed from a periodic unit cell, as shown in Fig. 2, with the geometrical dimensions summarized in Table I. The structure consists of a patterned resonator layer, a dielectric spacer, and a metallic ground plane arranged in a multilayer

configuration. The resonator geometry is defined by a circular ring embedded with uniformly distributed square slots, which are strategically designed to support strong localized resonances in the THz regime. The dielectric spacer separates the patterned resonator from the ground plane, while the continuous metallic layer at the bottom eliminates transmission and enhances absorption. The unit cell has lateral periodicities of $S_L=55 \mu\text{m}$ and $S_W=55 \mu\text{m}$. The outer ring radius of $r=25.2 \mu\text{m}$ with thickness of $c=0.33 \mu\text{m}$ and spacing of $d=1.05 \mu\text{m}$. The dielectric spacer thickness is $e=6.9 \mu\text{m}$, and the ground plane thickness is $f=0.22 \mu\text{m}$. Inside the ring, there are ten evenly spaced hollow square loops ($a=8 \mu\text{m}$) arranged in a circular pattern, forming a periodic radial array. This compact hybrid configuration enables multiple absorption peaks within the 0.1–2.5 THz range by concentrating the electromagnetic fields around the resonator features. The strong confinement of fields within and around the patterned square, ring slot geometry makes the absorber highly sensitive to variations in the surrounding dielectric environment. This ensures that any deposited analyte can be effectively detected through measurable resonance frequency shifts, enabling precise material characterization in the THz spectrum.

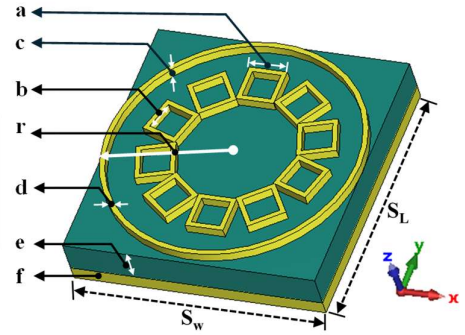


Fig. 2: Structural design of the proposed unit cell

TABLE I STRUCTURAL PARAMETERS FOR PROPOSED DESIGN

Parameter	Unit(μm)	Parameter	Unit(μm)
Substrate length, $S_L=S_W$	55	Ring width, d	1.05
Outer square arm, a	8	Substrate thickness, e	6.9
Inner square arm, b	6	Ground thickness, f	0.22
Metal thickness, c	0.33	Ring radius, r	25.2

A. Optimization of Parameters

To optimize the absorption response of the proposed terahertz (THz) metamaterial absorber, the influence of key geometrical parameters of the unit cell was systematically investigated. The study focused on the inner square arm length (b), outer ring radius (r), and ring width (d), while other parameters were kept constant (Table I). Fig. 3(a) shows the variation in ' b ' induced noticeable shifts in both resonance frequencies, with optimum absorption observed at $b = 6 \mu\text{m}$, where deep reflection minima indicate strong performance. Similarly, increasing ' r ' altered the current distribution along the resonator, producing the best absorption at $r = 25.2 \mu\text{m}$ with higher quality factors and enhanced confinement (Fig. 3b). Whether adjustments in ' d ' modulated the coupling between the incident THz field and the resonator, with $d = 1.05 \mu\text{m}$ yielding sharp, well-defined absorption dips. The

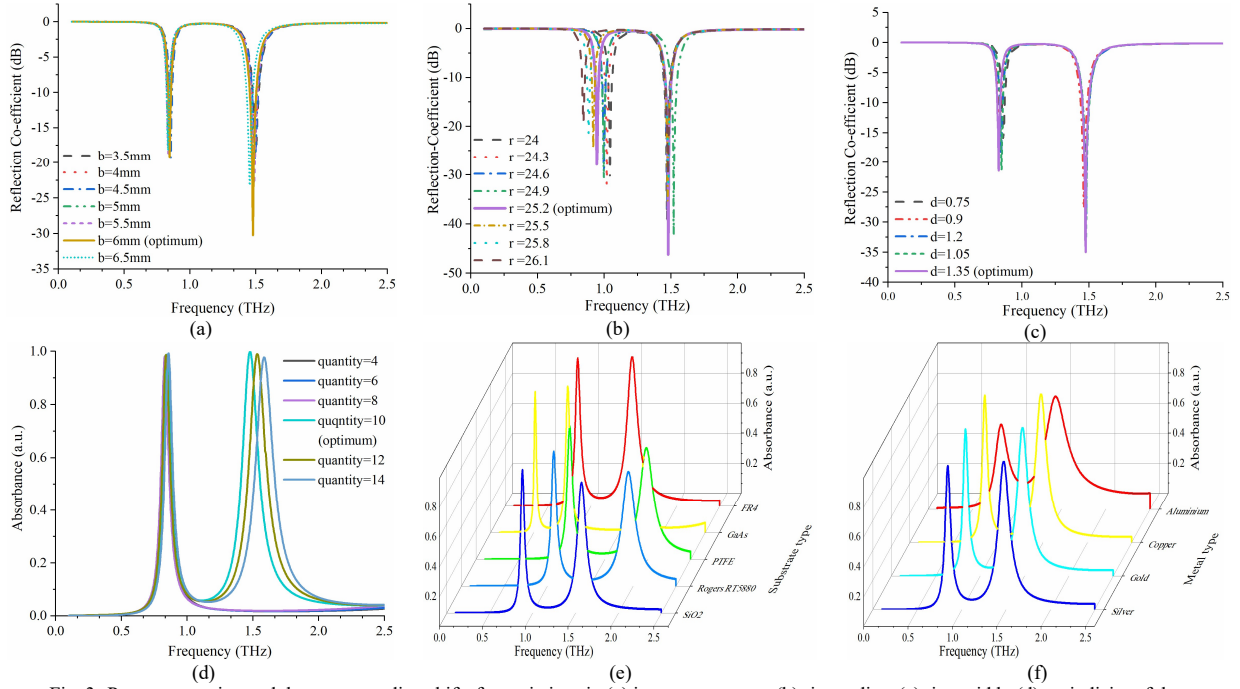


Fig. 3: Parameter tuning and the corresponding shifts for variations in (a) inner square arm, (b) ring radius, (c) ring width, (d) periodicity of the square resonator, (e) substrate, and (f) metal.

effect of resonator periodicity in Fig. 3(d) was examined by varying the number of square elements from 4 to 14. Results demonstrate that both resonance strength and frequency are highly periodic-dependent, with 10 elements producing optimal dual-band performance (>99% absorbance at 0.844 and 1.478 THz). This highlights the importance of precise periodicity control for maximizing absorption efficiency.

Substrate-metal combinations were also evaluated in Fig. 3(e) and Fig. 3(f) respectively. Among the tested configurations, FR4-copper achieved near-unity dual-band absorption at ~1.0 and 1.6 THz. High-permittivity FR4 enhances field confinement, while copper ensures strong plasmonic resonance. Other substrates such as GaAs provide sharp but narrow peaks, PTFE and Rogers RT5880 support low-loss broadband responses, and SiO₂/polyimide offer balanced performance. Comparisons of metallic resonator designs further confirmed that a complete copper patch on FR4 delivers optimal resonance, establishing it as the most effective configuration for broadband THz sensing and explosive detection applications. Fig. 4 compares the absorption behavior of the proposed THz metamaterial absorber under different excitation regions. The complete structure achieves dual resonance peaks at 0.84 and 1.478 THz with near-unity absorption, while the middle portion yields a single sharp resonance at 1.478 THz (99.99% absorption). The outside portion produces a weak, broad response (98.78%). These results confirm that integrated resonator geometry enables dual-band operation through enhanced electric-magnetic field coupling.

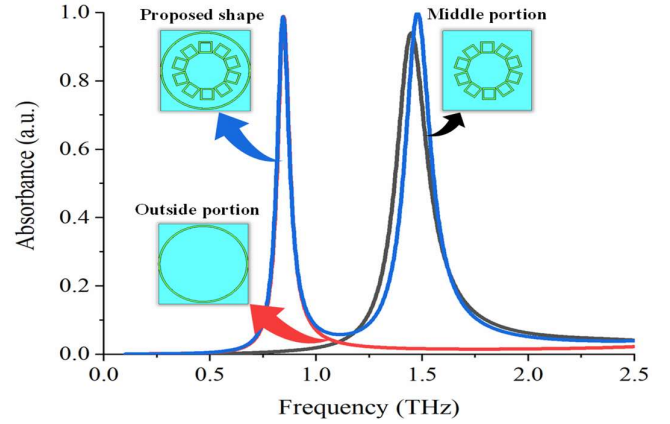


Fig. 4: Study of Absorption Behavior in Unit Cells in Different region

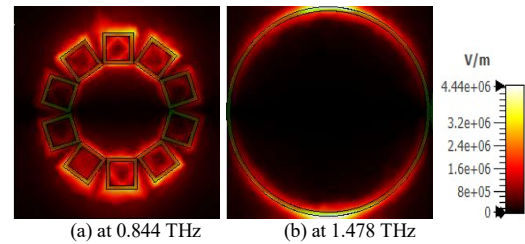


Fig. 5: E-field analysis for the proposed unit cell

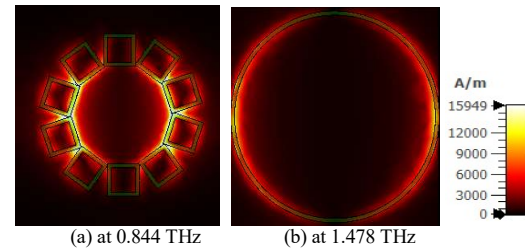


Fig. 6: H-field analysis for the proposed unit cell

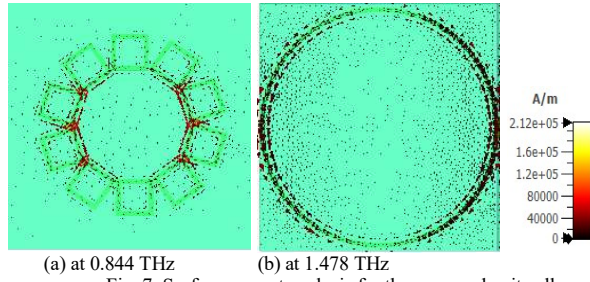


Fig. 7: Surface current analysis for the proposed unit cell

IV. RESULTS ANALYSIS

A. Field Distribution

The dual-band terahertz absorber exhibits distinct characteristics in electric and magnetic field responses at 0.844 THz and 1.478 THz (Fig. 4-7). At 0.844 THz, the E-field spreads vertically from the capacitive gaps toward the substrate (Fig. 5), producing strong electric resonance with a peak of 7.39×10^7 V/m due to capacitive coupling. Magnetic resonance in Fig. 6 dominates at 1.478 THz, with H-field 2.39×10^5 A/m extending horizontally from the center toward the edges and surface currents 3.08×10^4 A/m, arising from anti-parallel currents between the patterned surface and ground plane, enabling near-unity dual-band absorption depicted in Fig. 7.

B. Performance Analysis

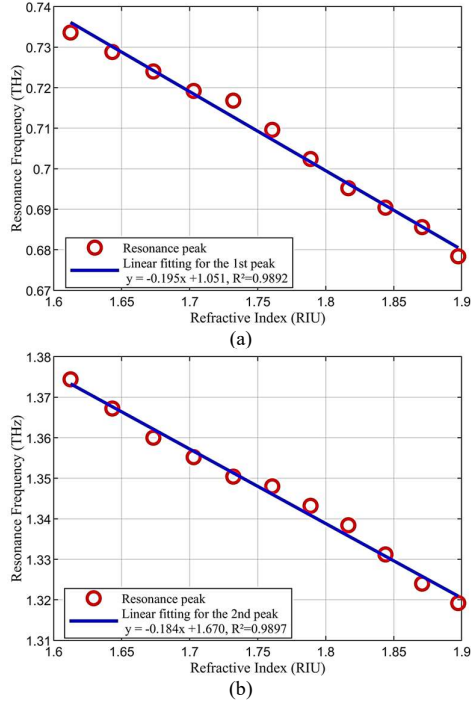


Fig. 8: Refractive index sensing performance of the unit cell with resonance frequency shifts for the (a) first and (b) second peak.

TABLE II UNIT CELL PERFORMANCE OVERVIEW

Parameter	1 st Peak	2 nd Peak	Parameter	1 st Peak	2 nd Peak
Resonance Peak (THz)	0.844	1.478	FoM	2.75	1.3
Sensitivity (GHz/RIU)	193.74	188.83	Q Factor	11.96	9.89
FWHM (THz)	0.071	0.149	Absorbance (%)	99.7	99.99

The unit cell performance with respect to refractive index (RI) for the proposed dual-band THz absorber is summarized under the two resonance peaks in Table II. Fig. 8(a) and 8(b) show the first and second peaks with sensitivities of 193.74 and 188.83 GHz/RIU, respectively, exhibiting a linear redshift and confirming reliable response. A trade-off is observed, as the first mode achieves a sharper FWHM (0.071 THz), higher FoM (2.75), and Q-factor (11.96), indicating superior detection precision. In contrast, the second mode provides complementary broadband sensing. Together, both modes validate the effectiveness of absorber as a compact, high-sensitivity platform for THz spectroscopy and explosive detection.

V. SENSING RESPONSE

The analytical performance of the proposed THz metamaterial absorber (MMA) was evaluated by tracking resonance frequency shifts for explosives with distinct characteristics of refractive indices (Fig. 9, Table III). Using air ($n = 1$) as the reference, the resonance at 0.8296 THz redshifted to 0.6924 THz for Tetryl ($n = 1.583$), 0.6468 THz for Semtex-H ($n = 1.645$), 0.6168 THz for TNT ($n = 1.6432$), 0.589 THz for HMX ($n = 1.75$), and 0.558 THz for RDX ($n = 1.772$). The corresponding frequency shifts (Δf) range from 137.2 GHz (Tetryl) to 271.6 GHz (RDX). Sensitivities of 315.78–351.81 GHz/RIU were achieved, confirming the absorber's strong responsiveness. The detection mechanism arises from near-field confinement of THz waves within the metamaterial resonator. When an explosive analyte with a higher refractive index is introduced in proximity to the resonator surface, it alters the effective dielectric environment, thereby increasing the effective optical path length and causing a shift of the resonance toward lower frequencies (redshift). This interaction is highly material-specific because each explosive has a characteristic refractive index, enabling selective identification. Importantly, the absorber maintains high efficiency across analytes, ensuring stable detection with minimal errors. The near-linear relation between resonance frequency and refractive index validates the sensing principle. Thus, the dual-band MMA absorber not only distinguishes between explosives such as TNT, Tetryl, Semtex-H, HMX, and RDX but also explains the physical basis of detection: resonance frequency shifts induced by refractive index variations. This establishes the THz metamaterial absorber as a robust and label-free detection platform for explosive identification and security applications.

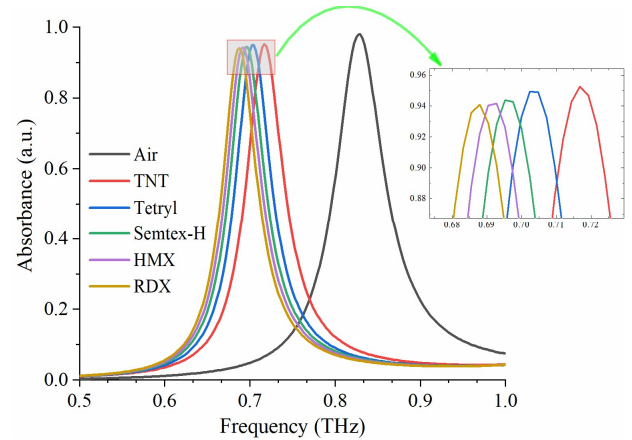


Fig. 9: Investigation of analytics changes associated with resonance frequency shifts in explosives

TABLE III DETECTION OF EXPLOSIVES AND SENSORS PERFORMANCE

Substance	Name of Explosive Chemical	RI [9]	f_r (THz)	Absorbance (%)	Δf (GHz)	Sensitivity (GHz/RIU)
Air	Air	1	0.8296	0.979	Ref.	Ref.
TNT	2,4,6-Trinitrotoluene	1.6432	0.6168	0.9526	212.8	330.85
Tetryl	2,4,6-Trinitrophenyl-N-methylnitramine	1.7029	0.6024	0.9493	227.2	323.23
Semtex-H	RDX-PETN	1.732	0.5952	0.9438	234.4	320.22
HMX	Cyclo tetramethylene tetranitramine	1.7499	0.5928	0.9416	236.8	315.78
RDX	Hexahydro-1,3,5-trinitro-1,3,5-triazine	1.772	0.558	0.941	271.6	351.81

VI. COMPARATIVE ANALYSIS

As detailed in Table IV, the proposed dual-band absorber demonstrates a superior balance of performance metrics for explosive detection. It achieves exceptional absorbance (>99.7%) alongside the highest Q-factor (11.96) among all listed works in Refs. [5-8], indicating superior resonance sharpness and spectral selectivity. Furthermore, it attains a

leading Figure of Merit (FoM of 2.75). While some designs report higher sensitivity, this work offers a superior balance of sensitivity (351.81 GHz/RIU) and resonance sharpness. Crucially, it is proven for a broader range of explosives (TNT, Tetryl, Semtex-H, HMX, RDX) compared to the more limited targets of other studies, establishing it as a versatile and high-performance sensor.

VII. CONCLUSION

This study introduced a dual-band terahertz metamaterial absorber tailored for selective explosive detection. By integrating a concentric ring with square slot resonators on an FR-4/copper platform, the absorber achieved near-unity absorption at 0.844 and 1.478 THz, supported by strong electric and magnetic field localization. The device demonstrated high refractive index sensitivity for RDX (up to 351.81 GHz/RIU), stable absorption across different analytes, and the capability to differentiate explosives such as TNT, Tetryl, Semtex-H, HMX, and RDX. Comparative evaluation confirmed that the proposed design outperforms existing absorbers in terms of Q-factor, Figure of Merit, and detection precision. Overall, the results validate the absorber's effectiveness as a compact, high-performance THz sensing platform for non-destructive explosive identification, offering strong potential for deployment in real-world security and defense systems.

TABLE IV COMPARATIVE ANALYSIS OF EXISTING LITERATURE

Ref.	Operating range (THz)	Material substrate	Q-Factor	FoM	Absorbance (%)	Sensitivity (GHz/RIU)	Application
Dhillon et al. [5]	0.00557, 0.00797, 0.01344	FR4	N/A	N/A	98.87, 97.99, 99.28	N/A	Defense, Explosive Detection, Radar
Barati Haghverdi et al. [6]	6	Kapton	N/A	N/A	>90% (peak)	N/A	Methane Gas Sensing
Sharma et al. [7]	0.8, 1.05, 1.3, 1.91	Silicon	N/A	N/A	>99.9%	N/A	RDX Explosive Detection
Wekalao et al. [8]	1.357, 1.379, 1.437	Glass	7.414 - 7.684	2.151 - 2.368	N/A	443	Explosive Detection (TNT, Nitro)
	1.224, 1.244, 1.295	Glass	6.442 - 6.582	0.805 - 4.905	N/A	927	Explosive Detection (TNT, Nitro)
This work	0.844, 1.478	FR4	11.96, 9.89	2.75, 1.30	99.7, 99.99	351.81	Explosive Detection (TNT, Tetryl-1, Semtex-H, HMX, RDX)

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